

Decision Tree Classifier with the Gini Coefficient

Gini Impurity Formula

$$\text{Gini} = 1 - \sum (p_i)^2$$

where p_i is the proportion of samples belonging to class i in that region. Gini = 0 means the region is pure (all one class).

Part 1: The Dataset

Below is a scatter plot with 20 samples. Each sample has two features (x, y) and belongs to one of two classes:

- ● Class A (red circles)
- ▲ Class B (blue triangles)

Training Data Scatter Plot

Draw your decision boundaries directly on the plot below. Use a ruler to draw straight vertical or horizontal lines.



Part 2: Calculate Gini Impurity

For each split you drew, calculate the Gini impurity of the resulting regions.

Split 1

Feature & Threshold	Left/Lower Region	Right/Upper Region
Split on: x = _____ or y = _____	# Class A: _____ # Class B: _____ Gini = _____	# Class A: _____ # Class B: _____ Gini = _____

Split 2

Feature & Threshold	Left/Lower Region	Right/Upper Region
Split on: x = _____ or y = _____	# Class A: _____ # Class B: _____ Gini = _____	# Class A: _____ # Class B: _____ Gini = _____

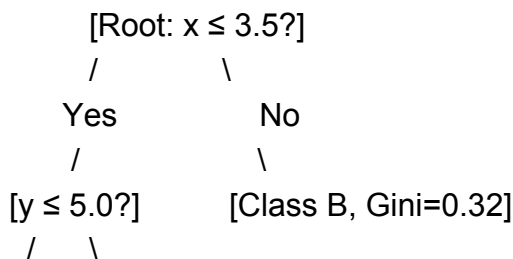
Split 3

Feature & Threshold	Left/Lower Region	Right/Upper Region
Split on: x = _____ or y = _____	# Class A: _____ # Class B: _____ Gini = _____	# Class A: _____ # Class B: _____ Gini = _____

Part 3: Draw Your Decision Tree

Draw the tree structure that corresponds to your decision boundaries. Each internal node should show the split condition, and each leaf should show the predicted class and Gini impurity.

Example format:



Part 4: Decision tree on three classes

Decision Tree Training Data (3 Classes)

